

Exercise No. 5

ASYMMETRIC CRYPTOGRAPHY

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Task 3.1: RSA demonstration tool experiments

- **Public key value (e, n):** [To be filled after CrypTool experiment]
- **Private key value (d, n):** [To be filled after CrypTool experiment]
- **Results:** [To be filled]

Task 3.2: RSA encryption with different block lengths

- **Ciphertext (block length 2):** [To be filled]
- **Ciphertext (block length 1):** [To be filled]
- **Conclusions:** Encrypting with a block length of 1 character essentially turns RSA into a monoalphabetic substitution cipher, which is insecure and vulnerable to frequency analysis. Using a longer block length increases security by hiding single-character frequencies.

Task 3.3: 1024-bit RSA key pair generation

- **Variables composing the public key:** Modulus (n) and Public Exponent (e)
- **Value of the exponent:** [To be filled]
- **Certificate information:** [To be filled]

Task 3.4: RSA encryption and decryption duration

- **Encryption duration:** [To be filled]
- **Decryption duration:** [To be filled]
- **Conclusions:** Decryption typically takes significantly longer than encryption in RSA. This is because the private exponent (d) is usually much larger than the public exponent (e), requiring more computational time for the modular exponentiation process.

Task 3.5: Digital signature embedding

- **Duration of signing:** [To be filled]
- **Hexadecimal analysis:** [To be filled]
- **Is it possible to separate signature and data?:** Yes, the signature is typically appended or prepended to the data file. In hexadecimal view, the signature block can be identified and separated from the original file data.

Task 3.6: Impact of errors on the digital signature

- **Signature verification after changing signature characters:** Verification fails. The signature is recognized as invalid.
- **Signature verification after changing signed data characters:** Verification fails. Even a single bit change in the data results in a completely different hash, making the signature invalid.
- **Conclusions:** Digital signatures ensure both data integrity and authenticity. Any modification to either the signature itself or the signed data will cause the verification process to fail, alerting the user of tampering.

Task 3.7: Factorization of module n and calculating d

- **Key 1:** $n = 31313$, $e = 4913$
 - $p = 173$, $q = 181$
 - $d = 17117$
 - **Private key:** ($d = 17117$, $n = 31313$)
- **Key 2:** $n = 63978486879527143858831415041$, $e = 17579$
 - **Fastest method:** [To be filled from CrypTool]
 - $p =$ [To be filled]
 - $q =$ [To be filled]
 - $d =$ [To be filled]
 - **Private key:** [To be filled]

Task 3.8: RSA Cryptoanalysis using factorization

- **Ciphertext decryption:** [To be filled after setting correct alphabet options in CrypTool]
- **Decrypted text:** [To be filled]

Task 3.9: Attack on short private key

- p : [To be filled]
- q : [To be filled]
- **Delta coefficient:** [To be filled]
- **Check on numbers from Task 3.1:** [To be filled]

Task 3.10: Advantages and disadvantages of asymmetric cryptography

- **Advantages:**
 - **Key Distribution:** Eliminates the need for secure exchange of a secret key, solving the primary key distribution problem of symmetric cryptography.
 - **Digital Signatures:** Enables non-repudiation and authentication, allowing users to verify the sender's identity and data integrity.
 - **Scalability:** Requires fewer keys overall for a network of users compared to symmetric systems (n pairs instead of $n(n-1)/2$ keys).

- **Disadvantages:**

- **Computational Cost:** Significantly slower and more computationally intensive than symmetric cryptography due to complex mathematics involving very large numbers.
- **Key Size:** Requires much larger key sizes (e.g., 2048-bit or 4096-bit) to achieve the same level of security as symmetric algorithms (e.g., 128-bit or 256-bit).
- **Vulnerability to Quantum Computing:** Algorithms like RSA and ECC are theoretically vulnerable to Shor's algorithm and could be broken by a sufficiently powerful quantum computer.